If A can be diagonalized as PDP^{-1}, is the power A^n diagonalizable? If so, how do we diagonalise this power?

Answer:

Yes, if a matrix **A** can be diagonalized as $\mathbf{PDP}^{-1}$, then any integer power of it, $\mathbf{A}^n$, is also diagonalizable.

The diagonalization of $\mathbf{A}^n$ is found by raising the diagonal matrix **D** to the nth power, while keeping the matrices **P** and $\mathbf{P}^{-1}$ the same. Specifically, if $\mathbf{A} = \mathbf{PDP}^{-1}$, then the diagonalization of $\mathbf{A}^n$ is:

$$\mathbf{A}^n = \mathbf{PD}^n\mathbf{P}^{-1}$$

Let's walk through why this is the case and how it works.

The logic relies on the fact that the term $\mathbf{P}^{-1}$P cancels out to become the identity matrix (I).

1. **Consider $\mathbf{A}^2$:** If we write out $\mathbf{A}^2$, we get: $\mathbf{A}^2 = \mathbf{A} \cdot \mathbf{A} = (\mathbf{PDP}^{-1})(\mathbf{PDP}^{-1})$. Using the associative property of matrix multiplication, we can regroup the terms in the middle: $\mathbf{A}^2 = \mathbf{PD}(\mathbf{P}^{-1}\mathbf{P})\mathbf{DP}^{-1}$. Since $\mathbf{P}^{-1}\mathbf{P} = \mathbf{I}$ (the identity matrix), this simplifies to: $\mathbf{A}^2 = \mathbf{PDIDP}^{-1} = \mathbf{PDDP}^{-1} = \mathbf{PD}^2\mathbf{P}^{-1}$ So, $\mathbf{A}^2$ is diagonalized by the matrix $\mathbf{D}^2$, which is simply the original diagonal matrix **D** with each of its diagonal elements squared.
2. **Generalizing to $\mathbf{A}^n$:** We can extend this pattern to any positive integer power n.
   $\mathbf{A}^n = \underbrace{\mathbf{A} \cdot \mathbf{A} \cdot \ldots \cdot \mathbf{A}}_{\text{n times}}$
   Substituting $\mathbf{A} = \mathbf{PDP}^{-1}$ for each term: $\mathbf{A}^n = (\mathbf{PDP}^{-1})(\mathbf{PDP}^{-1}) \ldots (\mathbf{PDP}^{-1})$
   All the interior $\mathbf{P}^{-1}\mathbf{P}$ pairs will cancel out to the identity matrix in a "telescoping" fashion:
   $\mathbf{A}^n = \mathbf{PD}(\mathbf{P}^{-1}\mathbf{P})\mathbf{D}(\mathbf{P}^{-1}\mathbf{P}) \ldots (\mathbf{P}^{-1}\mathbf{P})\mathbf{DP}^{-1}$ $\mathbf{A}^n = \mathbf{P}\underbrace{\mathbf{D} \cdot \mathbf{D} \cdot \ldots \cdot \mathbf{D}}_{\text{n times}}\mathbf{P}^{-1}$
   This leaves us with the final result: $\mathbf{A}^n = \mathbf{PD}^n\mathbf{P}^{-1}$

Given that $\mathbf{A} = \mathbf{PDP}^{-1}$, where **D** is a diagonal matrix of eigenvalues and **P** is the matrix of corresponding eigenvectors:

1. **The Eigenvector Matrix (P) Stays the Same:** The matrix **P** used to diagonalize $\mathbf{A}^n$ is the exact same matrix of eigenvectors used to diagonalize **A**. Its inverse, $\mathbf{P}^{-1}$, also remains the same.
2. **The Eigenvalue Matrix (D) is Raised to the Power:** The new diagonal matrix is $\mathbf{D}^n$. If the diagonal entries of **D** are the eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_k$, then the diagonal entries of $\mathbf{D}^n$ are $\lambda_1{}^n, \lambda_2{}^n, \ldots, \lambda_k{}^n$.

For example, if: $\mathbf{D} = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$ Then: $\mathbf{D}^n = \begin{pmatrix} \lambda_1^n & 0 \\ 0 & \lambda_2^n \end{pmatrix}$

This property is incredibly powerful because calculating $\mathbf{A}^n$ directly can be computationally expensive, especially for large n. However, calculating $\mathbf{D}^n$ is trivial—you just raise the individual diagonal elements to the power. Therefore, diagonalizing **A** provides a highly efficient shortcut for computing its powers.

In summary, $\mathbf{A}^n$ has the **same eigenvectors** as **A**, but its **eigenvalues** are the nth powers of the eigenvalues of **A**.